

CAIRNGORM SUMMIT, United Kingdom

WMO: 030650

Lat: 57.1163N

Lon: 3.6439W

Elev: 1245

StdP: 87.24

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-10.1	-9.2	-23.7	0.5	0.1	-17.8	0.9	0.9	37.7	-2.3	35.8	-1.3	12.7	40	1.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	4.3	15.2	10.2	13.6	9.6	12.1	9.4	12.1	13.2	11.1	12.2	10.2	11.4	8.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.7	9.9	12.3	10.6	9.3	11.3	9.6	8.7	10.5	37.9	13.1	35.3	12.2	33.0	11.5	18.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
41.2	34.2	30.0	-11.0	17.8	1.6	2.2	-12.1	19.4	-13.1	20.7	-14.0	21.9	-15.1	23.5
			-11.4	14.6	1.6	2.4	-12.6	16.4	-13.5	17.8	-14.4	19.1	-15.6	20.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.3	-3.0	-3.1	-2.4	0.1	2.3	4.5	7.1	6.5	4.9	2.0	-0.8	-2.5
	DBStd	4.95	2.92	3.76	3.59	3.61	4.26	3.43	3.14	2.81	3.06	3.04	3.08	3.41
	HDD10.0	3196	404	368	385	296	243	171	102	115	155	248	323	386
	HDD18.3	6204	663	601	643	546	497	416	347	366	402	506	573	644
	CDD10.0	33	0	0	0	0	4	5	13	7	3	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	14.6	16.5	15.4	15.2	14.2	11.7	11.6	11.7	12.8	14.3	17.0	17.2	18.2
Precipitation	PrecAvg	897	94	65	63	56	62	66	68	79	70	92	90	93
	PrecMax	1175	225	183	161	130	111	140	145	170	228	148	159	223
	PrecMin	624	17	12	11	9	20	24	24	28	23	22	24	21
	PrecStd	114	47	39	33	29	24	29	31	32	39	35	29	47
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.2	9.3	9.7	11.3	16.3	17.1	16.9	16.8	14.4	11.3	8.8	7.7
		MCWB	3.4	5.4	4.9	8.0	10.9	9.9	11.4	12.5	8.2	5.4	5.9	4.3
	2%	DB	4.5	6.6	6.5	8.5	13.6	14.0	14.9	13.7	12.3	9.1	6.6	5.6
		MCWB	3.3	3.5	2.5	5.2	9.2	8.4	10.1	11.2	8.2	6.6	4.0	3.9
	5%	DB	2.9	4.2	4.1	6.7	10.7	11.2	13.3	11.9	10.9	7.7	5.1	4.2
		MCWB	2.5	3.0	2.4	4.2	8.4	8.4	10.0	10.2	8.7	6.1	4.0	2.9
	10%	DB	1.4	2.3	2.6	5.2	8.4	9.3	11.9	10.6	9.5	6.5	3.7	2.7
		MCWB	0.8	1.7	1.6	3.4	6.4	7.2	9.9	9.3	8.1	5.1	2.8	1.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.5	8.0	6.1	9.8	12.2	12.3	13.2	14.1	11.6	9.2	7.0	6.6
		MCDB	6.1	8.2	7.3	9.8	14.0	14.5	14.0	15.9	12.2	9.6	7.4	6.8
	2%	WB	3.6	5.1	4.0	6.7	10.2	10.6	11.9	11.9	10.4	7.9	5.2	4.7
		MCDB	3.7	5.6	4.8	7.4	12.5	12.4	13.0	13.0	10.9	8.1	5.3	4.9
	5%	WB	2.2	3.3	2.8	5.0	8.6	8.9	11.1	10.7	9.2	6.8	3.9	3.3
		MCDB	2.4	4.0	3.4	5.9	10.7	10.4	12.3	11.6	10.0	7.2	4.1	3.6
	10%	WB	0.8	1.8	1.5	3.6	6.6	7.6	10.0	9.6	8.2	5.6	2.7	1.7
		MCDB	1.1	2.4	2.2	4.5	8.1	9.0	11.5	10.3	9.0	6.0	2.9	2.0
Mean Daily Temperature Range	5% DB	MDBR	3.8	3.9	4.1	4.4	4.7	4.3	4.3	3.9	4.0	3.9	3.8	4.0
		MCDBR	6.7	6.3	6.5	6.4	6.2	6.3	5.7	5.6	5.4	6.1	6.3	7.0
		MCWBR	6.1	5.9	5.6	5.1	5.1	5.5	4.6	4.8	4.9	5.5	5.9	6.3
	5% WB	MCDBR	6.3	6.3	6.3	5.6	6.1	5.6	5.3	5.3	5.1	5.9	5.8	6.8
		MCWBR	6.1	6.3	5.9	5.2	5.1	5.2	4.6	4.9	5.1	5.8	5.7	6.6
Clear-Sky Solar Irradiance	taub	0.250	0.271	0.291	0.306	0.309	0.309	0.311	0.301	0.290	0.279	0.254	0.232	
	taud	2.336	2.461	2.462	2.467	2.492	2.508	2.500	2.536	2.576	2.561	2.417	2.243	
	Ebn at Noon	665	813	882	916	931	933	923	913	878	796	661	568	
	Edh at Noon	45	62	80	93	97	97	96	86	71	56	41	36	
All-Sky Solar Radiation	RadAvg	0.45	1.09	2.11	3.47	4.65	4.57	4.42	3.49	2.45	1.30	0.58	0.32	
	RadStd	0.06	0.11	0.24	0.34	0.28	0.40	0.38	0.23	0.20	0.13	0.05	0.04	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (37 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page